Supplementary information for "Local gods, supernatural punishment, and the expansion of human sociality: A cross-cultural causal analysis (Preregistration)"

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1 Supplementary Methods

1.1 Description of Data Collection Phases

As noted in the main text, data were collected in three phases (summarised in Supplementary Table S1). The first phase was data from Wave I (Purzycki et al., 2024) of the Evolution of Religion and Morality project (ERM; 2014-2015), with Random Allocation Games (RAGs) played with eight societies. In Wave II (Lang et al., 2024) of this project (2017-2018), data from an additional 7 societies were collected who played both RAGs and Dictator Games (DGs; with 4 societies from Wave I re-visited and DGs played).

In these two earlier phases, participants were generally recruited based on their affiliation (actual or default) with the "moralistic" religious tradition, and the cooperative experimental games were played with local and distant co-religionists associated with the same moralistic tradition (other than some exceptions and edge-cases discussed below). For the majority of societies (10/15; 67%), the designated "moralistic god" was a moralistic high god of either Christianity (7 sites) or Hinduism (3 sites). The five exceptions were: Samburu, Kenya (a syncretic combination of both Christian and traditional gods); Turkana, Kenya (also a syncretic combination of both Christian and traditional gods); Tyva, Russia (Buddha); Inland Tanna, Vanuatu (a traditional indigenous god); and Hazda, Tanzania (also a traditional indigenous god). These previous waves of data collection have been described in detail elsewhere, including many aspects we do not focus on here, such as the use of primes and games played with outgroup members from Wave II (Purzycki et al., 2016b; Lang et al., 2019; Purzycki et al., 2016a).

The third phase of data collection occurred more recently in 2024-2026 as part of the "Gods, Games and the Socioecological Landscape" (GGSL) project, which included seven additional sites: Sihanaka (Madagascar), Coastal Benin, Buryat (Mongolia), Altai Uriankhai (Mongolia), BaYaka (Republic of Congo), Toraja (Indonesia), and Kichwa (Ecuador). Data collection was identical to the previous ERM project, with the exception that participants were recruited based on their affiliation with the local religious tradition, and the experimental games were played with distant co-religionists of the same local religious tradition.[1]

1.2 Description of Study Sites

For the ERM sites, these descriptions are largely taken and adapted from Purzycki et al. (2016b) and Lang et al. (2019).

[1]Note that in the case of Coastal Benin, there were more combinations across traditions (e.g., some Vodun participants played with Christian recipients and some Christian participants played with Vodun recipients). For comparability across sites, here we focus on the co-religionist dyads of Vodun and Christians (i.e., Vodun participants with Vodun recipients, or Christian participants with Christian recipients). See Table 1 in the main text.

Group	Phase	Economy	N	Games
Altai Uriankhai	GGSL	Herder	86	RAG and DG
BaYaka	GGSL	Forager	98	RAG and DG
Buryat	GGSL	Herder	82	RAG and DG
Cachoeira	ERM II	Market	274	RAG and DG
Coastal Benin (Christian)	GGSL	Fishing/Market	91	RAG and DG
Coastal Benin (Vodun)	GGSL	Fishing/Market	256	RAG and DG
Coastal Tanna	ERM I/II	Horticulture	178	RAG and DG
Hadza	ERM I/II	Forager	201	RAG and DG
Huatasani	ERM II	Horticulture	94	RAG and DG
Indo-Mauritians	ERM I/II	Market/Farming	245	RAG and DG
Inland Tanna	ERM I/II	Horticulture	112	RAG and DG
Kananga	ERM II	Market	200	RAG and DG
Kichwa	GGSL	Horticulture	85	RAG and DG
Lovu	ERM I	Market	76	RAG
Marajó	ERM I	Market	77	RAG
Mysore	ERM II	Market	165	RAG and DG
Samburu	ERM II	Herder	40	RAG and DG
Sihanaka	GGSL	Farming	79	RAG and DG
Sursurunga	ERM II	Horticulture	163	RAG and DG
Toraja	GGSL	Market	80	RAG and DG
Turkana	ERM II	Herder	247	RAG and DG
Tyva	ERM I	Market/Herder	81	RAG
Yasawa	ERM I	Fishing/Farming	75	RAG

Supplementary Table S1 Additional summary description of 22 study sites. ERM = Evolution of Religion and Morality Project; GGSL = Gods, Games and the Socioecological Landscape project; RAG = Random Allocation Game; DG = Dictator Game.

1.2.1 Altai Uriankhai, Mongolia

The Altai Uriankhai are traditionally nomadic herders who live in Western Mongolia in a sparsely-populated mountainous range on the eastern slopes of the east-central area of the Altai Mountains, with a current population of approximately 3,700 people. The majority of Altai Uriankhai households subsist through pastoralism, relying primarily on sheep and goats, with yaks playing mostly a secondary role. Many herding households also maintain small numbers of horses (typically between one and thirty), while cattle and camels are kept only occasionally and to a much lesser extent. Although herding is still the main mode of subsistence, they are increasingly integrating with a market-based economy, particularly in residential centres. Consistent with all known Central Eurasian pastoral societies, they live in a patriarchal (patrilineal) society with a patrilocal (virilocal) residence structure.

As is common throughout Mongolia and Inner Asia, the Altai Uriankhai religious landscape is a syncretic combination of Tibetan Buddhism and local animistic and (to a lesser extent) shamanistic beliefs and practices. Based on this, for the purposes of this study we designated Buddha – largely perceived as both omniscient and punitive – as our moralistic god, and "spirit-masters" as our local god(s). These spirit-masters are locally-bound deities associated with the environment; e.g., spirits of the mountains, water, sky, etc. While rituals towards these deities take many forms, here we focus specifically on practices and offerings dedicated to spirit-masters at ritual cairns (*ovoo*), the primary purpose of which is to seek blessings for good fortune, including protection from natural calamities such as heavy snowfall, drought, and other environmental hardships, as well as to pray for longevity and good health.

The specific field site for this project was in Bosgo *bag* (third-level subdivision) of Duut *sum* (administrative unit of a province) in Khovd *aimag* (province), Western Mongolia. All Altai Uriankhai in this area identify with these spirit-master traditions, and hence were eligible for participation in this study. Based on Mongolian population census data, the sample of participants was broadly representative of this area, in terms of age, sex, community size and birthplace. For the economic games, co-religionists were defined as "an anonymous *ovoo* practitioner" either from Duut *sum* (the local area; for local co-religionist recipients) or Bayandun *sum* (over 2,000km away in Eastern Mongolia; for distant co-religionist recipients). The interviews and the experiments were primarily conducted in a two-room outbuilding of a private home in Bosgo (80 of the total 88 participants), with a smaller number of sessions held in classrooms at Khovd State University in Khovd city (8 participants). Data collection occurred in Summer 2024.

1.2.2 BaYaka, Republic of Congo

The BaYaka are hunter-gatherers who belong to the Western cluster of the Congo Basin forager groups. As part of the Congo Basin, the area comprises dense lowland forests, swamps, and rivers, supporting the highest biodiversity in Africa. The BaYaka population in this region is estimated at between 24,000 and 48,000 individuals. Today, these forest cultures are separated by thousands of kilometers and speak different languages, but their foundational cultural schemas and institutions remain highly analogous, especially among groups where traditional subsistence profiles have persisted even amid eco-cultural change and state interventions. The BaYaka are prototypical "immediate-return" hunter-gatherers, where foraging subsistence strategies yield direct, immediate reward, and consumption occurs shortly after production. This present-oriented character runs through the whole body of the culture, where immediate action, spontaneity, and responsiveness are strongly emphasized over long-horizon planning, abstract reasoning, and pronounced institutional

control. Locally normative value commitments and beliefs are reinforced and reproduced within a highly prosocial, open and egalitarian environment, with no definite rules of private property or formal positions of authority.

The present-oriented dynamics of BaYaka life significantly shape the modes of knowledge acquisition, transfer, and storage, including that of religious knowledge. Knowledge is rarely externalized as explicit symbolic models or systemized conceptual frameworks. The culture makes little effort to stabilize an objective *body of teachings* through propositional claims independent of bodily engagement. Instead, everyday practices constitute the primary empirical setting where cosmological order is enacted and made visible. A central concept is that of *ekila*, a cultural force that regulates thought and action, developing ways of sharing food and managing human physical, mental, and reproductive potentials. In BaYaka thought and cosmology, there is little conceptual separation between nature and culture, or between *natural causality* and *moral order*, and the BaYaka coordinate the world as one continuous field of relations in which humans, animals, and spirits occupy distinct yet permeable positions. The forest creator god, *Komba*, from whom sharing roles are derived, was designated as the BaYaka's moralistic god. Compared to other deities, *Komba* was uniquely attributed with omniscience, in addition to rewarding people for their behaviour and being concerned with moral behaviours (note that few BaYaka rated any deities as "punitive"). We chose the collective *Mokondi* as the local deity. *Mokondi* are more specialised entities than *Komba*, and are tied to practices and characteristics culturally associated with gendered roles (e.g., hunting among males). The *Mokondi* system is a partially shared, partially individualized set of relations that are repeatedly mobilized to guide action – this repertoire can flexibly be utilized to draw on, assemble, and stabilize individual spirit-profiles based on individual preferences and context. Thus, we decided to use the umbrella term, *Mokondi*, to refer to forest spirits in general.

This research took place specifically among the Motaba BaYaka, with the field site located near the northern borders of the Republic of the Congo, in the Likouala Department. Data were collected from all BaYaka in one camp in the area. For the economic games, co-religionists were defined as "a BaYaka spirit-initiate" either from the same camp (for local co-religionist recipients) or from a distant camp (for distant co-religionist recipients). Data collection occurred in Summer 2025. As noted in more detail below, due to the abstract nature of the RAG protocols, these had to be adapted for the BaYaka context (i.e., physically touching one of the cups, rather than just thinking about it).

1.2.3 Buryat, Mongolia

The Buryats are an ethnic Mongolian people whose origins trace to the Khori and Tümed lineages inhabiting the northern regions of Mongolia, southern Siberia, and northeastern China. As a result of imperial conflicts and geopolitical reconfigurations over the centuries, the Buryats – derived from the Russian term *bratia* ("brother") – were divided among three modern states: Mongolia, Russia, and China. Most households depend on pastoral livestock herding as their main livelihood, sometimes supplemented with small household vegetable gardens during the summer. On average, households own between 150 and 800 animals, predominantly sheep and goats, but also cattle and horses. Wealthier herding families typically have larger numbers of sheep and goats, while smaller herds are more cattle-dominant.

As is common throughout Mongolia and Inner Asia, Buryat communities historically and contemporarily maintain a syncretic religious worldview that blends Buddhist cosmology with the veneration of spirit-masters (often referred to as "shamanism"). This dual religious orientation

reflects both the long-standing Buddhist influence introduced through Mongolian state formations and the resilience of local cosmological traditions grounded in kinship, ecology, and reciprocity with the natural world. We therefore designated Buddha – largely perceived as both omniscient and punitive – as our moralistic god, and spirit-masters as our local god(s). Shamanism, the beliefs in spirit-masters – locally known as *uul usnii ezed* ("lords of mountains and waters"), *tenger* or *tengri* ("sky"), and *ongod* ("spirits") – has existed for thousands of years, with many different ritual practices devoted to these locally-bound deities. Of particular relevance to this study are ritual practices and offerings dedicated to spirit-masters at ritual cairns (*ovoo*). These offerings to the local spirits are primarily for good fortune, such as avoiding environmental hardships (e.g., heavy snow, drought), health and longevity, and good luck when hunting wild game. Offerings typically include products such as milk, cheese, confectionery, alcohol and meat, among other items.

This research focuses on the *sum* (district) of Bayandun, which has a total population of 3,189 people in four bags (subdistricts: Türgen, Khairkhan, Yargai and Naran), and is located in the *aimag* (province) of Dornod in eastern Mongolia. Over 90% of the population self-identify as Buryat. All Buryat in this area identify with these spirit-master traditions, and hence were eligible for participation in this study. Participant selection aimed to achieve representation across *bags*, age groups, and gender. However, due to the environmental challenges and road conditions discussed above, the original target sample of approximately 20 participants per *bag* had to be revised, resulting in recruitment focusing on the more central and easy-to-access areas. For the economic games, co-religionists were defined as "an anonymous ovoo practitioner" either from Bayandun *sum* (the local area; for local co-religionist recipients) or Duut *sum* (over 2,000km away in Western Mongolia; for distant co-religionist recipients). On rainy days for data collection in the *sum* centre, the research team conducted interviews indoors at their rented accommodation, a classroom at the local primary school, and a room at the district hospital. For data collections outside of the *sum* center, all interviews were conducted in the participants' homes. Data collection occurred in Summer 2024.

1.2.4 Cachoiera, Brazil

Cachoeira (population 15,000) is located about 75 miles inland from the city of Salvador da Bahia in Northeastern Brazil. Nuclear and extended family households constitute the main societal units in Cachoeira, and female-headed households are common. Small-scale commerce, informal day labor, and tourism are the main sources of income. We chose the Roman Catholic God as the moralistic deity and *Ogum*, one of the *Candomblé orixás*, as the local god. *Candomblé* is a religion of the African Diaspora, which centers on the cultivation of *axé*, the life-force of the universe. *Olorun* or *Olódúmaré* is the creator god, but most rituals are directed at the *orixás* and other entities (including *Ogum*). The religion is organized around autonomous groups called *terreiros* led by a priestess or priest (*mãe* or *pai-de-santo*). While *Candomblé* norms and beliefs are distinct, there is some syncretism with Roman Catholicism and *Candomblé* adherents see little conflict between following the religion and believing in the Christian god or other deities.

In Cachoeira, participants were recruited based being *Candomblé*, while local co-religionists were fellow *Candomblé* members from Cachoeira. Distant co-religionists were defined as *Candomblé* practitioners in Salvador. The RAGs took place in January 2015. Testing was conducted in either local temples called *terreiros* as a local-god location prime or in secular areas, such as private homes and a community center. Participants were members of the specific *terreiros* or were recruited through word of mouth. The DGs were played in August 2015 at a community center with several

private rooms and waiting areas. Object primes were placed on top of the table where games were played and included a Christian bible (moralistic god prime), *Candomblé* beads (local god prime), and a police hat (secular prime). The control condition was an empty table. For more information on this site, see the site-specific report in Soler et al. (2022).

1.2.5 Coastal Benin

The communities in Benin are located in islands on the coastal lagoons of the southern part of the country, in the departments of Mono and Atlantique. They mostly rely economically on fishing (both as a subsistence activity and as a market activity), reed harvesting and processing, and salt mining. According to our estimates, the Mono community numbers around 570 individuals, with almost 90% of the population being constituted by Xwla or Xuedah speakers (closely related local languages). The Atlantique community is estimated to be around 960 individuals; as with the Mono community, Xwla/Xuedah speakers constitute almost 90% of the population, with a significant Fon minority (Fon being the predominant language in the southern part of Benin). The Atlantique community is more market-integrated, as it is better connected to urban areas.

The religious environment is rich and varied. Most interviewed individuals report Vodun as their primary religion (between 70-90%, depending on the community), with the rest self-reporting as Christians. Vodun is a traditional African religion with a pantheon of multiple deities, called *Vodun*, associated with a given role (e.g., *Ogun* is the god of metals, *Zangbeto* is the policing/punishing god) or a given ethnicity or lineage. The spiritual hierarchy of this religion, with a distant creator deity and a rich landscape of more proximate, local deities, allows many believers to interchangeably identify the Vodun creator god with the Christian god. The communities thus display a high level of religious syncretism. Several Vodun deities are perceived as holding strong concerns with people's moral behaviour. For example, most people interviewed for the Religious Landscape Interview agreed that deities such as *Heviosso* (god of thunder), *Ogun* (god of metals), and *Zangbeto* (policing god) are concerned with moral behaviour, while less agreement was found for *Dan* (the python god, strongly associated with the sacred city of Ouidah) and *Sakpata* (deity of smallpox). Given this religious landscape, the Christian god was selected as the moralistic god, with the collective "Vodun spirits" designated as the local god(s).

Interviews were conducted at individuals' homes in the communities by a team of field assistants. The target sample was any adult who was resident in the community at the time of interview. A first round of interviews (May-July 2025) collected economic games data from 707 individuals, while a second round of follow-up interviews (July-September 2025) was used to administer the demographic and religiosity questionnaires. The retention at the follow-up interview stage was quite high ($n = 671$). As the communities are religiously diverse, the identity of the local co-religionist in the games was conditional on the focal player's self-reported religious identity, while the identity of distant co-religionist was randomised (between Christian and Vodun) to collect data on all possible religious allocation dyads (Christian-Christian, Christian-Vodun, Vodun-Christian, Vodun-Vodun). Distant co-religionists were specified as being from the other community (e.g., from Mono department if the participant was from Atlantique department, or *vice versa*). The communities are far apart enough that we can exclude major contacts between their residents, although some personal ties may exist sporadically (e.g., the residents of either community are aware of the existence of the other).

1.2.6 Coastal and Inland Tanna, Vanuatu

Tanna is a tropical volcanic island situated at the southern end of a chain of approximately 65 inhabited islands that make up the Pacific nation of Vanuatu. Participants were sampled from two locations on Tanna: a predominantly *Kastom* Inland Tanna site and a predominantly Christian Coastal Tanna site. The Coastal Tanna site is a contiguous set of Christian coastal villages near the market town of Lenakel, with a total population of approximately 500. Residents are a combination of wage laborers, traders and subsistence farmers, some of whom have spent time living and working in Port Vila or doing seasonal work in Australia or New Zealand. All attend church but many also participate in local *Kastom* rituals. The Inland Tanna site is a rural *Kastom* community of subsistence farmers living in three small hamlets with a total population of approximately 90. In an effort to preserve the traditional *Kastom* way of life, individuals in these villages tend to eschew money and are much less likely to own electronic devices, or have worked for a wage.

Religion on Tanna is a mix of Christianity and traditional Melanesian and Polynesian beliefs and practices. Christian missionary activity began in the late 1830's following European contact and had a profound effect on the religious life of the island. Today, over 60% of the population of roughly 28,000 reports belonging to a Christian church. Despite the influence of Christianity, a 2009 census indicated that a fifth of the Tannese population identify as belonging to the *Kastom* religion. *Kastom* on Tanna is a revival and re-interpretation of pre-contact mythology and traditional cultural practices and encompasses beliefs in a number of indigenous gods and ancestor spirits spread across the island. One of the most powerful supernatural agents, *Kalbaben*, features in many folk stories and is often referred to as the creator god, taking up residence in the highest peak on Tanna, Mt. Tukosmera. *Tupunus* refers to the spirit or spiritual forces surrounding the garden, and by extension to the local magic practitioner who controls the spirit-force. Every evening, men (including many self-identified Christians) engage in ritualistic kava-drinking and associated libation to *Kalbaben*. Given the religious divide between Coastal and Inland Tanna, we chose the Christian God as the moralistic god and *Tupunus* as the local god at the Coastal site, and *Kalbaben* as the moralistic god and *Tupunus* as the local god at the Inland site.

Due to the small size of the *Kastom* communities, we were able to sample all adults from the three *Kastom* hamlets for the Inland sample. The Coastal participants were recruited via cluster sampling from a village census, with the goal of sampling at least one individual from each household. Participants at the Coastal site were exposed to one of three randomly allocated prime conditions for the duration of game testing: a neutral control, a Christian prime, or a *Kastom* prime. Due to sample size constraints and widely varying levels of exposure to Christianity at the *Kastom* site, *Kastom* participants were primed with only the *Kastom* prime or neutral control. The Christian prime consisted of a cloth under the game area with a drawing of a church, a bible and a crucifix necklace. The *Kastom* prime consisted of cloth with images of three items used in *Kastom* ritualistic dance: a ritual *toka* stick, a feather stick, and a bundle of dried grass known as a *wilgen*. The neutral prime consisted of cloth with an image of three painted flowers. Primes were placed in front of the participant on the game area before they arrived for testing and remained there until after they left the testing area, thereby ensuring that the participant played all the DGs under the same prime conditions. During gameplay, the cups were placed on top of the prime material. For more information on this site, see the site-specific reports in Atkinson (2018) and Vardy and Atkinson (2022).

1.2.7 Hadza, Tanzania

The Hadza are an indigenous population of hunter-gatherers living a semi-arid region at the base of a branch of the East African rift valley close to the Lake Eyasi basin. They live in small, mobile camps of approximately 30 individuals. Hadza life is characterized by high levels of food sharing, little to no stratification (e.g., chiefs), flexible residential patterns, monogamous marriage (although polygamy is occasionally practiced) and high levels of fertility. While their economy is largely subsistence-based – surviving on hunted game and gathered fruit, honey and nuts – the Hadza also frequently supplement their revenue with monetary earnings from tourists.

A particularly interesting feature of the Hadza, for the purposes of the present study, is that they were claimed to be minimally religious, as portrayed in past ethnographic descriptions. However, recent interviews with the Hadza suggest that a majority believe in a god or gods. In fact, an entity long featured in Hadza cosmology, *Haine*, may have acquired a god-like status to some Hadza, possibly due to the influence of Christian missionaries, and hence was selected as the moralistic god at this site. However, the majority of Hadza still did not know or did not believe in *Haine's* ability to enact supernatural feats. Another cosmic entity, *Ishoko* (the Hadza local god in Wave I of the ERM project), who the Hadza also named as a god, appears to be incorporated into Hadza beliefs about *Haine*. Most Hadza, when asked if *Haine* and *Ishoko* were the same or distinct entities, reported that they were one and the same. Hence, we selected only *Haine* as the moralistic god for Wave II, with no local god.

The local co-religionist was defined as a Hadza person living in the same camp as the participant, and the distant co-religionist represented another Hadza person living in a distant, unspecified camp. Because of the fluid and flexible living arrangements of the Hadza, distant co-religionists may have been or could be future in-group members.

The study used cups of maize instead of money, which is a more meaningful currency for the Hadza. The researchers first traveled to a camp located in an area that the Hadza have been known to occupy in past years. After establishing contact in the first camp, the Hadza in that location directed researchers to the next nearest site, and so on. For more information on this site, see the site-specific reports in Apicella (2018) and Stagnaro et al. (2022).

1.2.8 Huatasani, Peru

Huatasani, a district in the Southern Peruvian Altiplano in the department of Puno, is home to 4,100 people, about 2,000 of whom live in the town of Huatasani. The research was conducted both in this town and in some of the districts' rural communities during the dry season (July and August 2015). The primary mode of subsistence in the area is agro-pastoralism. Many people, particularly in town, complement agro-pastoralism with commerce in larger cities, raising domesticated animals for market, producing cheese, or mining in an area about a 3-hour bus ride north of town.

The vast majority of people in Huatasani consider themselves Catholic, and practice Catholicism in a largely syncretic form with influences from traditional Andean religious practices. *Dios*, the Christian God, was therefore selected as the moralistic god in our study. For the local god we first chose local spirits of the landscape called *apus* (from traditional Andean cosmovision). However, during the initial interviews many participants mentioned that they did not believe in *apus*. As such, 53% of participants were primarily asked in depth questions about their beliefs and practices regarding Catholic saints as local gods rather than *apus*. Specific Catholic saints and instantiations of the Virgin are often important for local communities, neighborhoods or areas.

Distant co-religionists were described as a Catholic person from either Vilque Chico or Taraco,

similarly sized districts about 1.5 hours away by car on the Aymara or Quechua side of the language boundary, respectively. Huatasani is primarily on the Quechua side of the Quechua-Aymara linguistic boundary. The town is about 75% Quechua-speaking, but some of its nearest communities are Aymara-speaking. The distant communities that we included in this study were both overwhelmingly Quechua-speaking, but lay on different sides of the language boundary.

Our sampling protocols differed in town and the rural communities. In the former, we primarily utilized a convenience sampling method after making a general invitation to participate during the town's neighborhood meetings. These were carried out over several weeks. In the rural communities, we made an announcement after community meetings and took down the names of interested parties whom we then called in sequence to participate over the course of one or two intense days of running the study. We also assigned half of the DG participants to a moralistic god prime, and the other half to a control condition. The moralistic god prime featured a picture of a local mountain with a cross on top. The cross was photo-shopped to be bigger than its actual size, and a decorative border of crosses was added to the image. The control condition was a picture of a non-local mountain with a decorative flower border. To ensure that the participants saw the primes, these images were placed on the trays on which the cups sat during gameplay.

1.2.9 Kananga, Democratic Republic of Congo

Kananga, a city of roughly 1 million (the fourth largest in D.R. Congo), is the seat of the Provincial Government of Kasai Central, one of the poorest provinces of the D.R. Congo. Kananga has a moderate climate, and is situated at the transition from the equatorial forest to the savannah. It is an administrative city, in which the Provincial Government is by far the largest employer. The great majority of individuals, however, engage in some form of petty commerce, capitalizing on small arbitrage opportunities that arise in a place where decaying infrastructure creates sizeable frictions to the free flow of goods and people. Kananga is ethnically diverse, with 64% of the population identifying as ethnically Luluwa, but with sizeable minorities of Luntu, Luba, Kuba, Lele, Tetela, Songe, Bindi, Chokwe and other ethnic groups.

The religious landscape of Kananga is variegated. Nearly 100% of the population identifies as Christian, with 34% belonging to Pentecostal/born-again churches (*églises de réveil*), 23% to Catholic churches, and 15% to Protestant churches. The moralistic god in this setting is therefore the Christian God. Alongside professed beliefs in the Christian God, individuals reported beliefs in a number of local deities and ancestral spirits. We first chose as the local god *Kadima*; however, since no participant in the initial interviews indicated believing in *Kadima*, we used ancestral spirits as the local god instead.

Because born-again Christian denominations have the largest following in Kananga, we sampled only born-again Christians and defined the local co-religionists as born-again Christian Luluwa. Distant co-religionists were defined as born-again Christians who were not Luluwa. However, it is worth noting that the born-again churches are heterogeneous. Most churches were started by a specific charismatic founder. As such, we might expect the distant co-religionist association to be relatively weaker in this context compared to other sites.

Sampling proceeded as follows: first, a short 10-minute screening survey of 3,343 households was conducted in 2014. A satellite map of the city was divided into several hundred polygons, which were randomly sampled to be visited by enumerators. Within polygons, enumerators then randomly sampled households by counting a certain number of houses to skip between visits ("choose every Xth house"), and selected a random person from the household to participate. Selected participants

completed all surveys and activities at their homes in the privacy of pop-up tents. We used object primes for the DG. Specifically, we made custom towels, which we spread over the area where the game was explained and demonstrated. On these towels were large color photos of (1) an image of the Bible (moralistic god prime), (2) an image of traditional masks (local god prime), (3) an image of flowers (neutral), and (4) an image of the national police logo (secular authority prime). Due to an inability to secure a sufficient number of coins at this site, we used envelopes and bills instead of cups and coins. For more information on this site, see the site-specific report in Kapepula et al. (2022).

1.2.10 Kichwa, Ecuador

The Kichwa are Ecuador's largest indigenous group, making up just under 7% of the population. Lowland Kichwa (population 50,000-100,000) inhabit the expansive, flat Amazonian rainforest to the east of the Andean highlands. Our field site, Yawari Urku (Yawari), is one of 17 constituent villages of Rukullakta, a Kichwa pueblo in Archidona Canton, Napo Province. Rukullakta villages generate profits from the production of cacao, coffee, and other sustainable cash crops to minimize the risk of oil exploitation. Most households maintain a *chagra*, or large garden, to grow fruit, raise chickens, and produce other food. Although modern consumer goods are easily available in Archidona or nearby Tena, the provincial capital, respondents reported growing around 50% of their own food on average, indicating limited market integration at the household level. Several thousand people live in Rukullakta. The populätion of Yawari village is approximately 200.

Yawari residents are all baptized Catholic, but local Christianity is highly syncretic with indigenous beliefs and practices. Traditional Kichwa spirituality revolves around *Pachamama*, a female nature deity or spirit linked with the rainforest, local mountains, and horticulture. *Guayusa*, a caffeinated beverage made from tree leaves, is consumed daily and plays an important role in ceremonial life. Another culturally important drink, *chicha*, is made from the *chonta* fruit, a savory rainforest tree product. The annual village *Anniversario* festival features abundant invocations of *Pachamama* alongside an elaborate fruit produce display, drumming and dancing, *guayusa* rituals, shamanistic cleansings, and electing a "queen" of the village. *Pachamama* is thus abundantly associated with fertility, growth, and nature. A Catholic priest visits each Sunday to say Mass in the local community center. Many Kichwa enthusiastically participate in pilgrimages to local Marian shrines. We therefore designated the Catholic God (God the Father or the Trinity) as the moralistic god and *Pachamama* as the local god.

Nearly all subjects with complete demographic data (85 of 86 total) lived in Yawari or another village of Rukullakta. We designated *Pachamama* believers from the Yawari/Rukullakta community as local co-religionist recipients for games, while *Pachamama* believers from Palimbe, a Kichwa village in Pastaza Province about 50 miles south of Yawari, were selected as the distant co-religionist participants. Because road transportation in Pastaza Province is poor, the two villages are about four hours apart by car. Palimbe is comparatively isolated in the Amazon rainforest, and no one in our sample in Yawari had ever heard of it or knew anyone who lived there. Most Yawari residents spontaneously associated people in Palimbe with Kichwa practices such as drinking *guayusa* or *chicha* or with *Pachamama* and/or shamanism, indicating that Palimbe residents were seen as co-religionists. Interviews were conducted in Spanish. Experiments were conducted in classrooms in the local school, and interviews were conducted in the local Yawari community center building. Data collection occurred in February and March 2025.

1.2.11 Lovu, Fiji

Indo-Fijians are a diaspora population brought to Fiji from India by the British as indentured workers. Wage labor is the primary source of income but Indo-Fijians also farm sugar cane. Religiously, Indo-Fijians are primarily Hindus and Muslims though some are Sikhs or Christian. The present sample includes Hindus from Lovu village on the island of Viti Levu. Participants largely claimed that all Hindu gods are different aspects of one single deity, *Bhagwan* and this deity was therefore selected as the moralistic deity for this study. As one could not be identified, no local deity was selected. In the experiments, the distant co-religionist was a Hindu living on Vanua Levu, the second largest island in Fiji. For the Indo-Fijian sample, this study was conducted in Fiji-Hindi and English. Participants came primarily from the villages of Lovu Seaside and Lovu HART. Some additional participants came from the nearby villages of Koro Pita and Drasa. They were contacted in person at their homes ahead of time and asked if they would like to participate. The Lovu research group obtained names and contact information from those who agreed. Though specific time slots were given to all participants ahead of time, almost no one showed up in their allotted time slot. Because of this, participants were taken whenever they showed up. Since every identifiable Hindu household in Lovu Seaside and Lovu HART were contacted, all participants from those villages were accepted. Only participants from Koro Pita and Drasa who had been previously contacted, or who showed up at the same time as those that had been previously contacted, were accepted. Participants from outside of Lovu who had been told about the experiment by their friends or family members after their friends and family members had participated were not allowed to participate. For more information on this site, see the site-specific report in Willard (2018).

1.2.12 Marajó, Brazil

At the mouth of the Amazon River lies Marajó Island, Brazil. Pesqueiro is a small fishing village on the east side of Marajó Island. Residents of Pesqueiro rely primarily on fish sales and tourism. Most residents are Catholic, although some are Evangelical Protestants. For this sample, the moralistic deity was the Christian God (*Deus*), and Our Lady of Nazareth (*Nossa Senhora de Nazaré*), the region's patron saint, served as the local deity. The distant co-religionist was a Christian from Rondon, a distant but familiar town in mainland Pará state. For residents of Pesqueiro, the study was conducted in Portuguese. Participants were sampled from the entire village. An up-to-date census of the entire population (total: 309; 92 families) was obtained and all adults were included for random selection. Individuals were approached in their homes and invited to take part in the study. If unavailable, an alternative was selected from a reserve list (also randomly generated). Thirty-four out of a total 128 individuals were unavailable on the scheduled date for the study, leaving a total of 94 scheduled participants, who were randomly assigned to conditions. Fourteen people did not show for their session. For more information on this site, see the site-specific report in Cohen et al. (2018).

1.2.13 Pointe Aux Piments, Mauritius

Mauritius is an island nation in the Indian Ocean, located on the Tropic of Capricorn and forming part of the Mascarene archipelago. It has a population of 1.3 million people, composed of a wide range of ethnic groups, including people of Asian, African, and European origin. Our field site, Pointe aux Piments, is a rural village on the Northwest coast with a population of 9,000, mostly employed in agriculture, fishing, tourism, and other services. Mauritian Hindus, who comprised

our sample for this study, have a patrilineal, patrilocal kinship system. Although most households consist of nuclear families, residential plots of land typically consist of multiple patrilineally-arranged households, where the entire extended family lives around a shared central yard.

The majority of Hindus in Pointe aux Piments are adherents of Sanatana Dharma, which is a mainstream form of traditional Hinduism originating from Northern India. The most important god among our participants, *Shiva*, served as the moralistic god at this site. *Shiva* is known to have benevolent as well as fearsome qualities (Destroyer and Transformer). Belief in spirits is widespread in Mauritius irrespective of religious affiliation. These beliefs are a syncretic mix of Christian and animistic concepts of African origin and are largely shared between ethnic and religious groups. These spirits are the souls of dead people, and can be benevolent or evil, depending on the circumstances of the person's death and the rituals performed for them. This notion of spirit was used as the local god. The distant co-religionist was a Hindu from La Gaulette, which is a small, distant village in the South.

Data collection for the RAG took place over two separate periods. The distant RAGs and interviews were conducted in 2013. New participants for the DGs were recruited in 2015. Contextual primes were used for the RAGs. A local Hindu temple was used for the religious condition, while for the neutral context a restaurant was used. For the DG, we used an apartment building, with three separate rooms as playing areas, each equipped with a different visual prime (an image printed on a tablecloth). The religious prime consisted of a *Shiva* lingam (non-anthropomorphic symbol for the god *Shiva*), accompanied by a cobra and a trident (both attributes of *Shiva*) and by the text "Lord Shiva." For the secular authority prime, we used the local police emblem, which includes a depiction of a dodo and a deer, accompanied by the words "Mauritius Police Force (MPF)." Finally, for the neutral condition we used a picture of two daisies and a butterfly, accompanied by with the word "Butterfly." Participants in the DG were recruited by local assistants off the streets of the larger area of Pointe aux Piments, and were randomly assigned to one of these three conditions. For more information on this site, see the site-specific reports in Xygalatas et al. (2018) and Kundtová Klocová et al. (2022).

1.2.14 Mysore, India

Mysore is the southernmost city of the Indian state Karnataka, and is situated between the Kaveri and Kabini rivers. The climate is tropical savanna and summer spans the months of March through June, followed by a rainy season from July through November, and winter from December through February. The District of Mysore consists of an urban region with over 900,000 residents, and a rural region with over 1.6 million residents. Residents have an industrial subsistence-based economy, with household industries, agricultural labor, and cultivation comprising the main sources of income. The current study focused on people located in the urban region of the Mysore District.

The three major religions in the region are Hinduism (87%), Islam (8.9%), and Buddhism (1.4%). *Shiva* is the main god of Hinduism in Mysore and was selected as the moralistic god. We used *Chamundeshwari*, who is considered to be a form of *Durga*, as the local goddess. In Shaivite Hinduism, which characterizes the majority of Hindus in South India, *Shiva* is believed to be responsible for creation, destruction, and maintenance. *Chamundeshwari* is portrayed as a ferocious tiger-riding warrior goddess, but is also viewed as a "mother goddess" who protects all residents of Mysore. The local co-religionist was defined as a randomly selected person in the participant's neighborhood. The distant co-religionist was defined as a randomly selected Hindu from the neighboring state of Tamil Nadu. The religious practices of Hindus in Tamil Nadu are

very similar to those of Hindus in Mysore.

Locations for games and interviews included a variety of indoor and outdoor settings. Since statues of deities and religious images are ubiquitous in South Indian homes, extra care was taken to select places without religious paraphernalia. In all cases, the games were conducted in a private area separate from the waiting room and interview room. Research assistants targeted eight to ten adults in their communities who used convenience and snowball sampling to recruit Hindu participants. Although this sampling strategy was not random, participants varied substantially by age, sex, and occupation. To prevent the risk of collusion, we did not allow people to participate if they: 1) showed up uninvited after the introduction of the study in a given session; or 2) were invited by participants who had already participated. Across both the RAGs and DGs, half of the participants were presented with a prime and the other half were not. The *Shiva* lingam was chosen as the moralistic god prime because it is considered one of the oldest idols used to worship Shiva. For more information on this site, see the site-specific report in Placek and Lightner (2022).

1.2.15 Samburu, Kenya

Samburu County is a semi-arid region of 20,000 square kilometers, located in north central Kenya. The population is primarily ethnic Samburu, although there are small numbers of other ethnic groups as well. Livestock herding, or pastoralism, is the main livelihood, although the Samburu people are increasingly diversifying their economic activities to include wage labor and trade (in livestock, forest products, food items, and other consumer items). In areas with higher and more reliable rainfall, some Samburu grow crops including maize, beans, potatoes, kale and other vegetables. However, cultivation is a high-risk activity given the frequency of droughts. Samburu society comprises eight sections, each divided into a number of clans, sub-clans and lineages. It is a polygynous, patrilineal and patrilocal society. There is quite a bit of variation in settlement structure and a trend toward smaller settlements.

Samburu indigenous religion is monotheistic, with belief focused on the god, *Nkai* who is understood to be omniscient and omnipresent, but is especially associated with high places like mountaintops. No other gods were mentioned by our informants. Catholic and Protestant missionary activities among the Samburu date back to the 1930s, and the Christian god is also referred to as *Nkai*. *Nkai* was therefore selected as the moralistic god in this study. However, there is considerable variation in religious self-identification and practice. For example, all informants named rituals related to male age-grade ceremonies, including male circumcision and a series of rituals that mark the progress of men from warrior to elder status. Only a few informants named Christian practices such as attending church and celebrating holidays such as Christmas and Easter as important religious rituals.

Distant co-religionists were identified as people who shared the participant's religious affiliation or beliefs but who lived in Poro, an area about 50 kilometers away from Mbaringon (our field site). The inhabitants of Poro are ethnically Samburu; however, they are primarily from another kinship section. There is no obvious tension or conflict between the people of Mbaringon and Poro, nor are there strong positive ties, aside from the fact that they belong to the same ethnic group. The RAGs were conducted in Mbaringon in July 2014, at Lesorogol's research center. Forty players were drawn from an existing random sample of 100 households. One adult member per household was invited to play the games. Separate areas were established for pre-game surveys (demography), game play, post-game interviews, and payments. No religious primes were used. We conducted all the games during a two-day period, and then asked people to come back for the religiosity interview

over the next two weeks.

1.2.16 Sihanaka, Madagascar

The Sihanaka, meaning "people of the marshes," number over half a million and are among Madagascar's oldest of the eighteen officially recognized ethnic groups. They inhabit the fertile Alaotra basin in the island's central northeast, a lacustrine landscape of marshes, extensive rice plains, and Madagascar's largest freshwater body, Lake Alaotra. This "granary of Madagascar" supports livelihoods based on irrigated riziculture, fishing, and the keeping of zebu cattle and domestic fowl such as geese and ducks.

Sihanaka culture and cosmology have developed in close symbiosis with the wetland ecology, with riziculture and ritual structured by the annual cycle of the rainy season (November–April) and the dry season (May–October). Sihanaka cosmology is grounded in an immanentist ontology in which all realms of existence participate in a single continuous cosmos animated by the divine power of *hasina*. Comparable to Oceanian *mana* or Indonesian *semangat*, *hasina* circulates through plants, water, and words, and can be accumulated, preserved, and transmitted through ritual action. Rather than dividing nature from supernature, the cosmos is conceived as a graded continuum of beings, places and objects endowed with different degrees of *hasina*, the sacred force that animates all life. For this study, we designated Ndrianampanjaka as the moralistic god. Ndrianampanjaka is understood to be the undying sovereign of the area surrounding Anororo – corresponding to his precolonial realm — and is strongly associated with social order, adherence to custom, and the protection of the community. We selected the *zazavavindrano* – "water people" who dwell in Lake Alaotra — as our local god. Renowned for their luminous beauty and long hair, they inhabit underwater villages with houses, fields, and zebus mirroring human life. Guardians of their watery domains, they reward purity, respect, and quiet ritual observance while punishing pollution, betrayal, and disorder with illness, infertility, or madness. Although Christianity was introduced in the nineteenth century and churches are now widespread, this immanentist cosmology endures, with new entities such as *Jesosy* (Jesus) and *Jehovah* (God, Yahweh) having been incorporated into Sihanaka cosmology as exceptionally powerful yet worldly *zanahary* (divine beings).

This study took place in the town of Anororo, adjacent to Lake Alaotra in the north of the Malagasy highlands. Participants were recruited through door-to-door recruitment in Anororo. Recruitment aimed to capture a broad cross-section of the local population while maintaining feasibility within the constraints of field-based experimental research. For the economic games, co-religionists were defined as "a person who cultivates relations with *zazavavindrano*" either from Anonoro (for local co-religionist recipients) or Ambohidava (a town across Lake Alaotra; for distant co-religionist recipients). To reduce the risk of collusion, coordinated responding, or information spillover about the experimental games, recruitment proceeded neighbourhood by neighbourhood. The research team worked intensively within one section of Anororo for several consecutive days before moving to another area on the opposite side of town. When individuals indicated prior knowledge of the games, or when bystanders began to gather during sessions, recruitment in that area was immediately discontinued. Each session was conducted in the participants' homes or, occasionally, at the research team's host family's house. Sessions were private, with no other persons present. Data collection occurred in Summer 2025.

1.2.17 Sursurunga, Papua New Guinea

There are approximately 4000 speakers of Sursurunga who live in 19 nucleated villages along the southeastern coast of New Ireland, Papua New Guinea. Most Sursurunga are subsistence horticulturalists whose gardens provide a tuber-based diet of taro, various varieties of sweet potato, and manioc. Store-bought supplements such as rice and canned fish accompany some meals. Income from cash crops such as copra and cacao have long been the primary source of money. Matrilineal descent in the form of named clans governs a number of features of Sursurunga life, including access to garden land, marriage, and the sequence of mortuary rituals conducted in honor of the deceased.

All Sursurunga speakers maintain at least a nominal commitment to Christianity. Wesleyan Methodist missionaries established a church at Nokon around 1925 and at Tekedan shortly thereafter. Like many other Pacific Island peoples, the Sursurunga have incorporated Christianity into their belief systems, redefining the origins of moieties, the array of unseen beings, and the norms of generosity and reciprocity as biblical. For almost all Sursurunga, the most prominent being is *Káláu*: God the Father in Christian Trinitarian thought. Hence, *Káláu* was chosen as the moralistic god. There are also a number of local spirits, many of whom cause illness, injury, or misfortune of some sort, including those guided to do so by sorcerers. We chose *sirmát* as the local god, a spirit that is independent of the control of sorcerers. Men occasionally "marry" a *sirmát*, which means, among other things, that they do not disclose her as the source of any good fortune they might have. *Sirmát*'s primary interest in human behavior is pragmatic: they do not like and are believed to try to punish human behavior that disturbs their tranquility.

The local co-religionist was a member of their clan. The distant co-religionist label represented individuals of the same Christian denomination (i.e., same church) from another clan. For small communities, subjects were handed a 3 x 5 cm card with a number on it and were told that on an upcoming day, a certain set of numbers would be called upon to participate in the games. Corralling was used for larger groups, with a local research assistant around in order to make sure that there was no collusion between participants. Priming was used in both the RAGs and DGs. For the RAGs with a moralistic prime, the experiments were conducted in a church building. For the DGs with a moralistic prime, the moralistic-god questions from our interview were asked prior to the DGs, and the rest of the interview was concluded later. In the local-god condition, the local god questions from our interview were asked just before playing the DGs. For more information on this site, see the site-specific report in Bolyanatz (2022).

1.2.18 Toraja, Indonesia

The Toraja are an indigenous highland group in the Tana Toraja and Toraja Utara regencies of South Sulawesi, Indonesia. Most identify as Christian (mainly Protestant), while about 4,000 individuals (∼1.5%) are officially registered as "Hindu" under Indonesia's recognized-religions framework and continue following their ancestors' autochthonous beliefs. In Simbuang District, these adherents are known as *Aluk to Dolo* (or *Alukta*) followers. They sustain themselves through rice farming for household and community use, coffee farming that supports their moderate market integration, and raise pigs and chickens central to ritual life. Both men and women contribute to work and care, with women more often cooking and men taking heavier labor and outside wage work. Post-marriage residence depends on family resources, and though children take the father's surname, ritual obligations extend to both parents' ancestors.

Aluk to Dolo is often confused with mainstream Hinduism. Though it shares the concept of plural nature deities and has felt some Balinese Hindu influence, *Alukta* has no written scriptures or

anthropomorphic idols; rituals pass down through descendants via local wisdom and ritual speech. The cosmos contains many sub-deities (*dewata*) tied to specific landscape elements – e.g., land, water, livestock – each invoked in context (planting, harvest, animal care, etc.). Especially important are ancestor spirits (*To lendu Membali Puang*), the realm which the souls of the dead transform and join to as *dewata* through proper funeral rites. They receive offerings and are called upon during funerals (along with the soul of the recently deceased), hardship, birth, harvests, and misfortune. Unexplainable events are typically attributed to ritual shortcomings and addressed with offerings of egg, rice, or pig, depending on severity and the local resources.

The two main ritual cycles are *Rambu solo′*, the elaborate "smoke descending" funerary rites sending the deceased to the afterlife, and *Rambu tuka′*, the "smoke ascending" rites of thanksgiving for harvests, births, and new houses. The two are kept ritually distinct so mourning and celebration never overlap. When someone dies, close family members often observe *mero′*, abstaining from rice and eating only corn until burial. The deceased is kept in the family home for months or years (depending on status and resources) before formal burial, with appropriate sacrifices at each stage. Reburial rituals may follow, involving many pig sacrifices and renewed invocation of ancestors. Many Toraja Christians also take part, so *Aluk to Dolo* remains woven into Toraja ceremonial life beyond its officially registered adherents.

For the economic games, *Puang Matua* was selected as the moralistic god – the name also used by local Christians for the Christian God. *To lendu Membali Puang*, the ancestral spirits, served as the local god. Our main field site was Puangbembe Mesakada village in Simbuang District. Co-religionists were defined as *Aluk to Dolo* adherents either in Puangbembe Mesakada village (for the "local" condition) or in a remote Mappak district village with same language and core rituals, but almost no contact with the field site (for the "distant" condition). Because participants had time and mobility restrictions, interviews and experiments were conducted in temporary locations across five central village areas. Data collection took place in Winter 2026.

1.2.19 Turkana, Kenya

The Turkana are semi-nomadic pastoralists with a population of approximately 1 million. They inhabit the semi-arid region of north-west Kenya, an area prone to frequent droughts. Within Kenya, the Turkana neighbor the Pokot, Samburu, Rendille, and Borana pastoral ethnic groups, with whom cattle raiding and interethnic tensions are common. They have low literacy rates and limited market integration. Most individuals make a living by herding cattle, small livestock (goats and sheep), and camels, which provide the main sources of household protein. Livestock are periodically sold in town to purchase staples like maize flour, beans, sugar, tea and tobacco. The Turkana are patrilineal, patrilocal, and polygynous. They are separated into approximately 18 different territorial sections (*ekitela*), which are geographically distinct, with each territorial section living on and having grazing rights to certain delineated land areas. Crosscutting these territorial sections are approximately 24 different ethnic "marks" or clans (*emacar*). Marriage is exogamous at the clan level.

The moralistic god in this study was *Akuj*, a primary deity among the Turkana who is associated with the sky, particularly as the giver of life and of rain (two elements that are highly associated in Turkana society). *Akuj* is said to live at the tops of mountains, especially those mountains that are associated with rain. He is benevolent and omnipotent, and actively controls all major blessings in life. With the expansion of Christian missions starting in the 1960s, Christianity co-opted *Akuj* as the creator and one true God, and took the Turkana term for "an evil spirit," known as *ekipe*, as a

single Satan character. There is considerable individual variation in the extent to which people have adopted Christian practices. When individuals who have been strongly influenced by Christianity refer to *Akuj*, they do not typically distinguish between the local primary god, *Akuj*, and the Christian monotheistic god, *Akuj*. Aside from *Akuj*, there is also a belief in the existence of small, terrestrial ancestral spirits known as *ngipean* or *ngikaram*, who are more limited in their scope and powers than *Akuj*. Some spirits can be quite malevolent and may require animal sacrifices to be appeased when they are angry. We chose these ancestral spirits as the local god.

The distant co-religionist was defined as "a Turkana person you don't know who attends the church in Awar-Naparan." Awar-Naparan is a Turkana settlement about 25 kilometers from the field site, moving closer to the town of Kakuma, and is about 2-3 times larger than the settlement areas used in this study. Given the distance to Awar-Naparan, it is unlikely that participants would have close/daily interactions with people there. The majority of the study participants and the people from Awar-Naparan belong to the same territorial section (Ngiyapakuno), and there are no recent or historical animosities between these two communities.

Prior to the start of this project, we had conducted a house-to-house census for the four settlement areas included in the study and had randomized the collected names for each location. The first 25 people from each list were used in the RAG sample, with each of the four locations corresponding to a different day of play. The next 50 names on each list made up the DG sample, with each location corresponding to a different day of play. The games were played in small tents and we used tokens (dried beans) instead of coins, as we were unable to locate a sufficient number of small coins near the field site. During the RAGs, we used two secular tents (each containing a bottle of water and notebook) and two moralistic god tents (each containing a bible, rosary, and large crucifix). The DGs consisted of four tents: one secular, one moralistic god, one local god (containing a clay jar with red ochre powder, which is commonly used by local seers for divination), and one secular authority tent (containing a camouflage jacket commonly worn by police officers).

1.2.20 Tyva, Russia

Hailed informally as the geographic centre of Asia, the Tyva Republic lies in southern Siberia, just north of the western portion of Mongolia. Urban Tyvans subsist primarily on a market-based economy while rural Tyvans herd sheep, goats, cattle, and/or yaks, This sample was drawn exclusively from the capital city of Kyzyl. Most Tyvans identify as Buddhist, but also engage in religious practices associated with shamanism, animism, and totemism. Buddha-Burgan ("Buddha God") functioned as the moralistic deity, while an unspecified *cher eezi*, or "master of the place," a spiritual lord over resources and regions functioned as the local deity. The distant co-religionist was from Ak Dovurak, a familiar asbestos-mining town about a 4-hours west of Kyzyl by car. All experiments and interviews were conducted in Tyvan, though some did ask for game instructions in Russian for clarity. The Tyva research group's efforts to have recruits participate in follow-up sessions were futile. They therefore conducted single sessions, each lasting around 90 minutes per participant. Four assistants used random, chain and snowball sampling to recruit people who would contact the lead assistant to coordinate meeting places and times. Assistants only divulged that they required up to 90 minutes of participants' time and that they would be paid for it. They also encouraged enlisted participants to also recruit more people before their participation, but not after, and they refused all unsolicited candidates. Assistants also asked each participant about all of the information that they knew about the study and everyone conveyed only the allowed information. Assistants recruited people on the basis of their Buddhist and/or Shamanist identification (note that

participants who identified as Shamanist were also Buddhist, meaning that all Tyvan participants were Buddhist), Tyvan ethnicity, and fluency of the Tyvan language. For more information on this site, see the site-specific report in Purzycki and Kulundary (2018).

1.2.21 Yasawa, Fiji

Yasawa Island is on the northwestern corner of the Fijian archipelago. Yasawans are primarily fisher-horticulturalists. The majority of Yasawans identify as Wesleyan Methodists though a large minority associate with the Evangelical Assemblies of God. However, traditional beliefs and practices devoted to ancestor spirits (*Kalou-vu* or "root/ancestor god") continue to thrive. For this sample, the moralistic deity was the Christian "Bible God," while the *Kalou-vu* represented the local deities. Cups for the distant coreligionists in Yasawa were Fijian Christians from another island. The Yasawan protocols were all conducted in Bauan. Indigenous Fijian participants were recruited by invitation based upon their location within the village. The games were played in houses across the village, one on each day. Villagers living closest to those houses were invited to attend in waves of eight participants by Indigenous Fijian research assistants who administered the games and post-game interviews. For more information on this site, see the site-specific report in McNamara and Henrich (2018).

1.3 Site-Specific Deviations from Core Protocol

On the whole, protocols were largely followed across all field sites. There were, however, some exceptions and complications in the data to note here:

- *Sursurunga:* As discussed above, across all sites the aim was for the religious affiliation of the participant to be aligned with the identity of the local and distant co-religionist recipients in the games (e.g., Christian participants with Christian recipients). However, this correspondence was not met among the Sursurunga, Papua New Guinea. In this site, participants were Christian, the local recipient was "a person from your clan" (i.e., conflating kinship and religious affiliation), while the distant recipient was "a person from your church" (i.e., a local, rather than distant, co-religionist). For this reason, we will exclude Sursurunga from all analyses reported here. This was the only site where this alignment was not met (see Table 1 in main text).

- *BaYaka:* When piloting the experimental games, the BaYaka had trouble with the abstract nature of the RAG instructions. Instead of just thinking of a cup/recipient, BaYaka participants were asked to place their hand on the cup of their choice, then roll a die with a hand image on three sides, with the other three sides blank. After rolling the die, players were supposed to put the token in the cup they were touching if the hand icon camp up, and the other cup if not. While the rules of the game were identical to the original RAG protocols, it is possible that participants may have altered their behaviour as others could potentially see their choices – as opposed to the standard RAG protocols where these decisions were purely mental and hence invisible to others (Hruschka et al., 2014) – even though games were played privately. However, based on ethnographic observations and post-game discussions with the BaYaka, we believe this is unlikely to unduly influence BaYaka game behaviour. Many BaYaka participants also had issues answering the Likert scale questions due to their often abstract or hypothetical nature; the local field assistants often had to change the wording or add clarifications in a way for participants to link the questions to context and local

experience. Where any responses were ambiguous, the response option nearest as possible to their answer was selected.

- *Hadza:* The Hadza had difficulty answering many of the questions on their original scale, particularly for Likert scale responses. Instead, to facilitate comprehension many religiosity questions were converted to trichotomous responses, with "Yes", "No" and "I don't know" as response options (Apicella, 2018; Purzycki et al., 2016b). As we predominantly focus on the binary "punishment" and "omniscience" variables here, this deviation among the Hadza is less of an issue compared to the Likert scale questions. However, it is unclear how to treat these "don't know" responses for comparability with the other sites. We will explore a range of options: i) removing these "don't know" responses; ii) combining "don't know" and "no" responses together; and iii) coding "don't know" responses as missing and using imputation techniques to impute plausible values on the binary scale. Additionally, as noted in the ethnographic description above, although *Haine* and *Ishoko* were used as the moralistic and local gods (respectively) for Wave I of the ERM, from follow-up interviews most Hadza reported that these were the same deities, hence there was no local god for the Hadza in Wave II. The food security time-frames also differed for the Hadza; with only "next month" and "next year" used in Wave I, and only "next month" used in Wave II (instead of "next month", "next six months", "next year" and "next five years" as with the rest of the sites).

- *Inland Tanna:* Some Inland Tanna participants also responded "don't know" to some of these binary religiosity questions about punishment and omniscience. The same methods as with the Hadza to deal with these responses will also be used for these data.

- *Huatsani:* As noted in the ethnographic section above, the local god changed mid-way through data collection. Initially local spirits (*apus*) were selected, but this was subsequently changed to Catholic saints as many participants said they did not believe in *apus*. Comparisons between moralistic and local gods in this site therefore need to consider this change in deity.

- *Lovu* and *Samburu:* In both of these sites there were no local deities, only the focal moralistic gods.

- *Altai Uriankhai:* Due to errors in the initial wave of data collection, questions involving scales did not use the Likert scale responses as intended. While participants' answers were matched to the closest response option where possible, there is uncertainty in these responses with many data-points missing. Given this, in the following field season in 2025 the religiosity interview was asked to the participants again using the correct Likert response scales; this was collected for ∼75% of the original participants. As this issue only affects Likert scale questions – and the majority of our analyses rely on binary variables, which were unaffected by this error – for the majority of analyses we use variables from the initial wave of data collection in 2024. The only exception is the "moral concern scale," used to assess our designations of "moralistic" vs. "local" gods (see Supplementary Section 1.4, below), which will use this newer 2025 data.

- *Sihanaka:* Among the Sihanaka, ideas surrounding the afterlife are cosmologically vague and are something which cannot be influenced by either Ndrianampanjaka or *zazavavindrano*, the Sihanakan moralistic and local deities used in the present study, respectively. Instead, of key importance upon death among the Sihanaka is entering the family tomb with the ancestors, which involves different deities to those assessed here. As such, many Sihanaka did not answer

the question "Can [deity] influence what happens to people after they die?", and of those that did, few answered "yes." Given this, for the Sihanaka we will apply three different coding options: i) focus only on the question "Can [deity] ever punish people for their behaviour?" as our proxy for supernatural punishment, rather than a punishment mini-scale of both these questions as used for all other sites; and ii) recoding these missing data as "no"; and iii) imputing these missing data.

1.4 Additional Discussion on Coding Moralistic vs. Local Religious Traditions

As discussed in the main text, we will assess three different operationalisations of moralistic vs. local religious traditions to examine if/how this alters any subsequent conclusions. These decisions are summarised in Supplementary Table S2. As can be seen, most of this uncertainty centers around Cachoeira, Hadza, Inland Tanna, Samburu and Turkana, while the designation of moralistic vs. local religious tradition is arguably less ambiguous for the other sites.

First, we will use a definition based on whether the focal religious tradition belongs to a religion such as Christianity, Buddhism and Hinduism, or not. We label these as "world religions" here, simply as a tag to refer to these religious traditions, without making strong claims regarding their history, origin or specific features which may (or may not) differentiate them from other religious traditions. It is simply a useful – but imperfect – label for these traditions which are often believed to be more "moralistic" than other religious traditions; readers can substitute this term with an alternative, such as "axial," "moral," "prosocial" or "big god" religion (see Bellah, 2011; Baumard and Boyer, 2013; Norenzayan, 2013; Norenzayan et al., 2016), if they prefer. That is, moralistic religious traditions will be defined as "any world religion," with local religious traditions as "any non-world religion." By this definition, local religious traditions would encompass many "indigenous" religious traditions, even if it is the most moralistic in said society. For instance, among the Hadza and Inland Tanna, the deities chosen as "moralistic" were traditional indigenous gods (*Haine* and *Kalpanan*, respectively), and clearly are not part of a world religious tradition. The Samburu context complicates this dichotomy, as their moralistic god, *Nkai*, refers both to a traditional indigenous god and a missionary-introduced Christian god, with many traditional – rather than Christian – rituals performed in relation to this tradition, and variation in adherence to Christianity within this society (see site description, above). The Turkana case is similar, as they combine local/indigenous deities with a Christian god, again with variation in adherence between Turkana individuals. As it is not clear whether these Samburu and Turkana religious traditions should be categorised as moralistic (world) or local (non-world), we will compare both classifications and assess whether this impacts our results. By this definition, either ten societies have local religious traditions with 12 moralistic ones (if Samburu and Turkana are coded as moralistic/world), or 12 societies have local religious traditions with 10 moralistic ones (if Samburu and Turkana are coded as local/non-world).

Next, we will adopt a "relative" (i.e., within-site) operationalisation of moralistic god traditions as "the more punitive, omniscient and morally concerned of the two traditions within a society" and local god traditions as "the less punitive, omniscient and morally concerned of the two traditions within a society." This follows the assumptions made in previous publications using ERM data (Purzycki et al., 2016b; Lang et al., 2019) that all these societies have participants and recipients from the more moralistic of the two religious traditions, while the seven new GGSL societies are from the local religious tradition (other than Benin, which also has data from a moralistic tradition as well). The only exception is Cachoeira, collected as part of the ERM project, where participants

and recipients were from the local *Candomblé* religious tradition, so will be categorised as local here. Note also that, for the two sites without a designated local god – Lovu and Samburu – we will classify these as "moralistic" traditions under this relative coding scheme. Based on this operationalisation, there are 14 societies with data from a moralistic tradition, and eight local ones. In addition to simply declaring these sites as being from either the "more" or "less" moralistic tradition, we will use the following data to confirm these designations are accurate (and update our categorisations, if necessary): punishment mini-scales, omniscience mini-scales, moral concern scales (i.e., how concerned each deity is with punishing lying, theft and murder), and free-list data on each deity's moral concern. That is, based on this "relative" definition, we expect the chosen moralistic god to be more punitive, knowledgeable, and morally-concerned than the local god within each society.

Finally, we will use an "absolute" (i.e., between-site) definition, based on a data-driven approach to determine whether the religious tradition is moralistic or local. For this, we will use free-list data of what angers/displeases the deity of the same religious tradition as the game co-religionists to understand how morally concerned these gods are, and categorise accordingly. For instance, based on previous publications the moralistic deities from Hadza and Inland Tanna seem less morally concerned than the moralistic gods from other societies (see Extended Data Figure 2 of Purzycki et al., 2016b). Consequently, we may code these sites as local, rather than moralistic, religious traditions. Conversely, although local gods for each society were ideally selected to be less morally concerned than moralistic gods, if a local god from one of the sites where participants and recipients are from the same local god tradition is very concerned with morality, we would categorise this as a moralistic religious tradition. Without seeing the full data we cannot make these decisions *a priori*, but will clearly communicate these in our final manuscript. As there will be variation in moral concern, we inevitably will have to make binary categorisations based on continuous data; we will, however, group these societies based on sensible clusters/cut-points in the data, and assess how different choices of categorisation (e.g., cut-points between moralistic and local traditions) may impact our results. Note that for this "absolute" definition we use moral concern data from free-lists, rather than the moral concern scales, for a few reasons. First, from a pragmatic perspective, many societies had difficulties collecting this "moral concern scale" data which was measured on a Likert scale. This means that this data was not easily understood by participants (e.g., among the BaYaka), a different response scale was used (e.g., among the Hadza), or there were other issues collecting said Likert scale data (e.g., among the Altai Uriankhai). This moral concern scale data therefore excludes some sites and/or makes comparisons between sites more difficult; in contrast, there were fewer noted issues with participants answering the free-list questions across societies. Additionally, we believe free-list data are more valid as they provide a more direct "window" into participants' cognitive representations and beliefs (Purzycki, 2025), so use this here rather than the moral concern scales. This is especially so as interpretation of the scale data are somewhat ambiguous and inconsistent between groups and tradition types. For example, this moral concern scale has little relationship with free-listed moral concerns (Bendixen and Purzycki, 2025), potentially because they tap into different reasoning processes; i.e., free-list data may reflect cultural models, while scale items may reflect more general reasoning about supernatural agents (Purzycki et al., 2022).

Group	Phase	Focal religious tradition	"World" definition 1	"World" definition 2	"Relative" definition 1 (researcher)	"Relative" definition 2 (data-driven)	"Absolute" definition (data-driven)
Altai Uriankhai	GGSL	Spirit-masters	Local	Local	Local	TBD	TBD
Buryat	GGSL	Spirit-masters	Local	Local	Local	TBD	TBD
BaYaka	GGSL	Forest spirits	Local	Local	Local	TBD	TBD
Cachoeira	ERM II	Candomblé	Local	Local	Local	TBD	TBD
Coastal Benin (Christian)	GGSL	Christianity	Moralistic	Moralistic	Moralistic	TBD	TBD
Coastal Benin (Vodun)	GGSL	Vodun	Local	Local	Local	TBD	TBD
Coastal Tanna	ERM I/II	Christianity	Moralistic	Moralistic	Moralistic	TBD	TBD
Hadza	ERM I/II	Indigenous	Local	Local	Moralistic	TBD	TBD
Huatasani	ERM II	Christianity	Moralistic	Moralistic	Moralistic	TBD	TBD
Inland Tanna	ERM I/II	Kastom	Local	Local	Moralistic	TBD	TBD
Kananga	ERM II	Christianity	Moralistic	Moralistic	Moralistic	TBD	TBD
Kichwa	GGSL	Pachamama	Local	Local	Local	TBD	TBD
Lovu	ERM I	Hinduism	Moralistic	Moralistic	Moralistic	TBD	TBD
Marajó	ERM I	Christianity	Moralistic	Moralistic	Moralistic	TBD	TBD
Indo-Mauritians	ERM I/II	Hinduism	Moralistic	Moralistic	Moralistic	TBD	TBD
Mysore	ERM II	Hinduism	Moralistic	Moralistic	Moralistic	TBD	TBD
Samburu	ERM II	Christianity/Traditional	Moralistic	Local	Moralistic	TBD	TBD
Sihanaka	GGSL	Water spirits	Local	Local	Local	TBD	TBD
Sursurunga	ERM II	Christianity	NA	NA	NA	NA	NA
Toraja	GGSL	Ancestor spirits	Local	Local	Local	TBD	TBD
Turkana	ERM II	Christianity/Traditional	Moralistic	Local	Moralistic	TBD	TBD
Tyva	ERM I	Buddhism	Moralistic	Moralistic	Moralistic	TBD	TBD
Yasawa	ERM I	Christianity	Moralistic	Moralistic	Moralistic	TBD	TBD

Supplementary Table S2 Summary of different operationalisations of moralistic vs. local religious traditions across study sites. Note that Sursurunga will not be included in these analyses due to irregularities in how co-religionists were chosen (see Supplementary Section 1.3), hence being coded as "NA." ERM = Evolution of Religion and Morality Project; GGSL = Gods, Games and the Socioecological Landscape project; NA = Not Applicable; TBD = To Be Determined.

1.5 Additional Discussion on Causal Assumptions

A key assumption for inferring causality from observational data is that of "(conditional) exchangeability" (Hernán and Robins, 2020); that is, there are no unmeasured confounds or open biasing paths between our target estimate of interest (our "estimand"; Lundberg et al., 2021). Graphical tools such as Directed Acyclic Graphs (DAGs) are a simple and convenient way of visualising, exploring and communicating these assumptions (Pearl et al., 2016). A central insight of this causal approach is that causality cannot be inferred solely from statistical relationships, but instead requires an *a priori* causal model to inform and interpret our statistical analysis (D'Agostino McGowan et al., 2024). This is a more principled approach where causal assumptions precede analyses, especially when compared to many previous cross-cultural studies in this area where the rationale for the choice of covariates included over a range of different model specifications was unclear, with little regard for causal considerations (e.g., Purzycki et al., 2016b; Lang et al., 2019).

In section 1.5.1, we expand upon the brief discussion in the main text regarding the assumptions embedded within our core DAG (Figure 2 in main text). After this, in section 1.5.2, we discuss additional assumptions required for a causal interpretation, beyond those embedded explicitly in our DAG.

Together, this is a long list of very strong assumptions, most of which we know are likely to be violated (at least to some extent) and which we cannot empirically test using observed data. Despite this, there are many tangible benefits to taking an explicitly causal approach when working with observational data (Rohrer, 2018; Hernán and Robins, 2020; McElreath, 2020). For starters, theories make causal claims, so we ought be clear that – for most research questions – our aim is to try and infer causal relationships. Second, a formal causal inference approach gives us clear criteria for when we can and cannot infer causality from observational data; this gives us a clear target to aim for. Even if we end up wide of the mark, at least it is clear what our intention was. Relatedly, by being open and honest about our limitations, we can clearly communicate to the wider research community how future work can build upon and improve our methods to try and get closer to that elusive causal target. By being clear about the assumptions necessary to infer causation, it forces us to actively engage with and confront the (often many) limitations of our research, hopefully avoiding the worst of scientific exaggeration and hyperbole. Finally, a thorough consideration of the causal structure of our data feeds through and informs our approach to statistical analyses.

1.5.1 Causal Assumptions Embedded Within our Core DAG

First, our exposure "supernatural beliefs" is the joint effect of both beliefs regarding punishment and omniscience of the focal deity (i.e., either the moralistic or local god). As data are cross-sectional and the causal relations between punishment and omniscience likely complex, we focus here on the *joint* causal effect of punishment and omniscience. That is, we do not know whether punishment beliefs cause omniscience beliefs, omniscience beliefs cause punishment beliefs, whether there is reciprocal causation, or whether they are caused by a latent unobserved factor (e.g., "religious vigilance", or other cognitive factors which may predispose individuals towards or away from religious beliefs). By modelling the joint causal effect of punishment and omniscience we largely side-step these thorny issues, but at the expense of not being able to estimate the causal effect of punishment and omniscience beliefs *separately*; this may also impact our ability to assume "causal consistency" in our causal estimates (see below). Our focal causal contrast will therefore be when punishment and omniscience mini-scales both take the values of "0" (no punishment/omniscience) vs. both "1" (maximally punitive/omniscient).

Second, a key aspect of our estimand is whether the effect of the exposure above is moderated by moralistic vs. local religious tradition; that is, the effect of supernatural punishment and omniscience beliefs on religious cooperation may vary by whether participants are from a local or moralistic religious tradition. This is represented in the DAG by the "Effect modifier" node (note that, as DAGs are non-parametric, there is no universal way to denote effect modification or interactions in DAGs). Operationalisation of this has been discussed in detail above.

Third, there are a key set of "core confounders" which – based on theoretical and empirical literature – we believe may plausibly cause both the exposure and the outcome (Baimel et al., 2022; Purzycki and Bendixen, 2024; Vardy et al., 2022; Lang et al., 2026). These consist of age, sex, education and food security. We believe our assumptions that these variables are likely to only be potential confounders is valid for several reasons: i) nothing can cause age or sex, meaning they can only possibly be confounders; ii) as this study was conducted on adults, education is likely to be completed, and hence fixed, for the majority of participants, meaning there is a reduced probability that education could be caused by either current religious beliefs or cooperation (although prior religious beliefs or cooperation, or that of parents, acting as an unmeasured confounder cannot be ruled out); and iii) while we cannot rule out religious beliefs or cooperative tendencies causing food security, we believe that food security is more likely to be a cause than a consequence of these variables.

Fourth, again due to the cross-sectional nature of the data there is ambiguity as to whether some variables are confounders, mediators, or both (or neither; grouped under the "Confounders and/or mediators" node). For instance, it is plausible that "number of children" may cause supernatural beliefs (e.g., people with more children become more religious), in which case this is plausibly a confounder. Alternatively, it is also plausible that religious beliefs impact people's fertility (Shaver et al., 2019), in which case "number of children" could be a mediator on the path from supernatural beliefs to religious cooperation. Reciprocal causation in both directions is also possible, hence the bidirectional arrows in Figure 2 in the main text. Similar concerns also occur for participation in religious rituals and beliefs regarding other deities (i.e., are they a cause, or a consequence, of supernatural belief towards the focal god?). For unbiased causal estimates, we need to adjust for confounders, but not mediators (van Zwieten et al., 2022), hence this situation poses a dilemma, as both adjusting and not adjusting for these variables could lead to bias. One solution is to explore how these different causal assumptions impact results, to see how stable (or not) conclusions are (Major-Smith, 2023). For example, if our focal causal contrast does not change dramatically by these different causal assumptions, we can have more confidence in inferring a potential causal interpretation.

Fifth, as with all observational data there is the risk of unmeasured confounders (denoted by the node "U" in the DAG) biasing our estimates. Unmeasured factors such as personality, genetics, or prior religious beliefs or cooperation may cause both the exposure and the outcome, which we cannot rule out. Unmeasured group-level confounding between sites could also result in bias in cross-cultural data such as these; however, as described in the main text, we use a "Mundlak" model to remove this biasing threat (Mundlak, 1978; Bendixen and Purzycki, 2026).

Sixth, although not confounders, there are a range of features of the experimental games which may impact cooperation (e.g., order of games, level of comprehension for RAGs); we will include these in our model as they may help improve precision in our estimand (at the very least, their inclusion will not introduce bias).

In addition to these assumptions regarding confounding and exchangability, there are a number of other assumptions required for a causal estimate to be unbiased, including no reverse causality,

no selection bias, no measurement error, positivity, causal consistency and no interference. We discuss these additional assumption in more detail in Supplementary Section 1.5.2, below.

1.5.2 Additional Causal Assumptions

The above discussion predominantly focused on assumptions regarding confounding and exchangability required for causal inferences, as summarised in the Figure 2 DAG in the main text. Next, we discuss a number of additional assumptions required for a causal estimate to be unbiased (Hernán and Robins, 2020; Bulbulia, 2024). These are:

1. No reverse causality. That is, cooperation with distant co-religionists does not cause supernatural beliefs. While cooperation in these games cannot directly cause religious beliefs, game behaviour is plausibly a proxy for prior cooperative behaviour or cooperative dispositions *outside* of the game context, which could cause supernatural beliefs (i.e., prior cooperation may be an unmeasured confounder). While we believe that the more probable direction is supernatural beliefs causing cooperation, from cross-sectional data we cannot formally rule out the possibility that broader aspects of cooperation also cause beliefs (e.g., cooperative individuals socially integrate better into religious traditions, bolstering their subsequent beliefs).

2. No selection bias. We have to assume that factors related to participation are not associated with both the exposure and outcome; i.e., there is no selection/collider bias (Hernán and Robins, 2020; Lu et al., 2022). Many data collections used random sampling, or sampled whole villages, so are hopefully representative of the target populations, especially in terms of religious beliefs and cooperation. Aside from the issue of sample-to-population representativeness, we also have to consider potential selection bias from the recruited sample to the analytic sample due to missing data (Hughes et al., 2019; Lu et al., 2022). While levels of missing data are generally quite low (we anticipate <10% missing data overall), this could result in bias in complete-case analyses if this missingness relates to our exposure and/or outcome variables; e.g., if both less religious and less cooperative individuals are more likely to have missing data. We discuss techniques to impute missing data and hopefully remove this bias in the "Statistical Models" section of the main text.

3. No measurement error. A further assumption is that we have to believe that individuals accurately report both their religious beliefs and their levels of cooperation, as such measurement error could result in bias (Lash et al., 2021). The RAG and DG are explicitly designed to try and minimise the risk of demand characteristics (e.g., anonymity, inability of researchers to be sure of any cheating [at least for the RAGs]), hopefully lowering the risk of bias. Relatedly, many fieldworkers involved in this project have extensive experience with their field-site (or fieldwork more generally), building up trust and hopefully lowering the risk of socially-desirable answers. However, we cannot rule out bias due to measurement error, which could range from the relatively benign (if measurement error is random, then we could get bias towards the null) to the more serious (e.g., if religious individuals over-exaggerate their willingness to cooperate, we could observe a spurious positive relationship between beliefs and cooperation).

4. Positivity. A further assumption for valid causal inferences is that of "positivity," meaning that for all levels of the exposure there is a non-zero probability of covariate levels/values. For instance, if all men believed that deities punish but no women did, this is a positivity

violation as no women would have data regarding belief in punishment (to give a simple example). The causal effect would therefore be biased, as it would be undefined/impossible to estimate among the strata of "women." This is one of the only causal assumptions we can assess empirically, and so will explore this on a site-by-site basis. To assess this, we will explore the distributions of covariates across both punishment and omniscience mini-scales using the R package "cobalt" (Greifer, 2025). The degree to which these covariate distributions overlap for each level of the exposure can be used to infer potential violations of the positivity assumption.

5. Causal consistency. This is the assumption that the observed exposure is identical (or similar enough) to a well-defined counter-factual "target trial" exposure. As different versions of the exposure can have different causal effects, it is important that the observed exposure closely aligns with a (potentially hypothetical) intervention (VanderWeele and Hernan, 2013; Hernán, 2016; Hernán and Robins, 2016). For our study, we are therefore assuming that the observed supernatural punishment and omniscience is comparable to if we were to intervene and manually determine these values. Whether this assumption is met is complicated by several factors. For one, it is not clear what the hypothetical well-defined intervention on supernatural punishment and omniscience would be. There are many ways deities can punish and monitor peoples' behaviour (How frequently would they punish? How would they punish? What would they punish for? What is their knowledge breadth? etc.). Relatedly, in reality individuals both within and between societies – and especially by religious traditions – will vary regarding their beliefs about the gods' punishment and omniscience, again meaning the exposure may not be particularly well-defined. Unfortunately, this situation is virtually unavoidable in cross-cultural datasets – and social science endeavours more broadly – with coarse measures, so we have to simply acknowledge that the consistency assumption may not be well-approximated here. Even so, there is little causal work in this area and we have to start somewhere, and hopefully future theoretical and empirical work can build on this to make more specific and well-defined exposures which better meet the consistency assumption.

6. No interference. A final causal assumption is that of no interference, meaning that one participant's exposure values do not influence others. As religion is a complex socially- and culturally-mediated set of beliefs and behaviours, it seems implausible that a change in one participant's religiosity will not affect others (especially in small groups). However, we again have to assume independence, even if it is not especially reasonable in reality. Conversely, a change in one person's religious beliefs will not automatically influence others, so the extent to which this assumption is violated – and the impact this would have on our causal estimate – is unclear.

7. Same data-generating mechanism in all sites. More broadly, as this is a cross-cultural analysis using multiple societies in the same analysis model, the assumption of (conditional) exchangability is even harder to meet than for research in a single society. One key reason for this is that both the confounders and direction of causation between variables could vary across societies. For instance, "number of children" or "punishment and omniscience beliefs regard other deities in a society" may be confounders in one society but mediators in another. However, we believe our proposed "core confounder" variables at least cover the basic core set of potential confounders likely to be present across the majority of societies. Of course, we cannot rule out these assumptions being violated for some sites, which would bias our both our pooled and some site-specific causal estimates.

1.6 Additional Information on Statistical Models

1.6.1 Structure of Bayesian Models

Below is the structure of the core statistical model for RAGs using a multi-level aggregated binomial model with fully-varying random effects for all variables measured at the individual level. Note that this is a "Mundlak" model, as it contains the mean punishment and omniscience values for each site as group-level covariates (i.e., the $\overline{pun}$ and $\overline{omni}$ fixed-effects), in order to account for potential group-level confounding of our focal estimand.

$$
\begin{aligned}
y_i &\sim \text{Binomial}(30, p_i) \\
\text{logit}(p_i) = \alpha_{\text{group[i]}} &+ \beta^{pun}_{\text{group[i]}} * \text{pun}_i + \beta^{\overline{pun}} * \overline{pun} \\
&+ \beta^{omni}_{\text{group[i]}} * \text{omni}_i + \beta^{\overline{omni}} * \overline{omni} \\
&+ \beta^{age}_{\text{group[i]}} * \text{age}_i + \beta^{sex}_{\text{group[i]}} * \text{sex}_i \\
&+ \beta^{edu}_{\text{group[i]}} * \text{edu}_i + \beta^{food}_{\text{group[i]}} * \text{food}_i \\
&+ \beta^{order}_{\text{group[i]}} * \text{order}_i + \beta^{comp}_{\text{group[i]}} * \text{comp}_i
\end{aligned}
$$

As noted in the main text, fixed-effects will be assigned weakly-regularising priors. The random-effects structure for group-level parameters is given by the following common variance-covariance matrix:

$$
\begin{bmatrix} \alpha_{\text{group}} \\ \beta^{pun}_{\text{group}} \\ \beta^{omni}_{\text{group}} \\ \beta^{age}_{\text{group}} \\ \beta^{sex}_{\text{group}} \\ \beta^{edu}_{\text{group}} \\ \beta^{food}_{\text{group}} \\ \beta^{order}_{\text{group}} \\ \beta^{comp}_{\text{group}} \end{bmatrix} \sim \text{MVNormal}\left(\begin{bmatrix} 0 \\ \beta^{pun} \\ \beta^{omni} \\ \beta^{age} \\ \beta^{sex} \\ \beta^{edu} \\ \beta^{food} \\ \beta^{order} \\ \beta^{comp} \end{bmatrix}, \mathbf{SRS} \right)
$$

$\mathbf{S}$ is a diagonal matrix of standard deviations of the intercept and all covariates of interest:

$$
\mathbf{S} = \begin{bmatrix}
\sigma_\alpha & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \sigma_{\beta^{pun}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \sigma_{\beta^{omni}} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & \sigma_{\beta^{age}} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \sigma_{\beta^{sex}} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \sigma_{\beta^{edu}} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \sigma_{\beta^{food}} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \sigma_{\beta^{order}} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \sigma_{\beta^{comp}}
\end{bmatrix}
$$

R is a full rank correlation matrix of the same variables. Each standard deviation is assigned an independent prior, and **R** is given a prior from the LKJ family (McElreath, 2020).

As noted in the main text, we will analyse the model's posterior predictions by local vs. moralistic tradition to examine potential effect modification. We believe this approach is more parsimonious than including "moralistic vs. local" tradition as an explicit interaction term for a few reasons: i) it avoids including more interaction terms (e.g., between the exposures and the effect modifier), reducing model complexity; ii) it avoids the inclusion of group-level fixed-effects with relatively few observations and hence lower variation; and iii) by using a fully-varying random effects approach these tradition-level differences are implicitly modelled in the differences between societies.

The DG models take the same structure as above, except that an ordered categorical model will be used instead of an aggregated binomial model. The model and random-effects structure are the same in both; the only differences are that: i) there will be no "comprehension" covariate, as these checks were only asked for the RAGs; and ii) since the intercept thresholds for ordinal models cannot be modelled as random effects in currently-available standard software, we will allow these parameters to vary by site as fixed-effects in the model.

1.6.2 Robustness Checks to Different Causal Assumptions

1. Different operational definitions of moralistic vs. local religious traditions: This is simple to assess, as the core model is the same for all operationalisations; the only difference is in the allocation of societies to either the moralistic or local religious tradition as discussed above and in the main text, which can be achieved during the g-computation phase with the posterior model predictions.

2. Multiplicative exposures. The base model assumes that the supernatural punishment and omniscience variables are additive (i.e., independent). As their relationship with cooperation could be synergistic – i.e., belief in punishment and omniscience together may have a larger effect than each independently – we will also explore whether this relationship is multiplicative by including an interaction between punishment and ominiscience. Leave-one-out information criteria (LOOIC) will be used to assess whether the inclusion of said interaction term improves model fit.

3. For variables designated as potential "confounders and/or mediators" in Figure 2 in the main text – beliefs about other gods, religious rituals towards the focal deity, and number of children – we will assess whether assuming they are confounders (i.e., included as covariates in statistical models) vs. mediators (i.e., not included as covariates in statistical models) alters the estimate of our focal estimand.

1.6.3 Additional Information on Missing Data and Imputation Methods

Based on our previous experience with these data, there are some variables with small amounts of missing data across most of the sites. Given the small amount of missing data, we anticipate that the risk of selection bias due to missing data is small, but possible (e.g., if both less cooperative and less religious individuals are more likely to have missing data). In addition, this missing data may result in less efficient analyses (i.e., wider uncertainty intervals) in complete-case analyses. We will therefore conduct multiple imputation to impute said missing data, boost statistical power, and minimise the risk of selection bias (if present). To avoid the complications of multi-level imputation and allow the imputation model to vary by site, we will impute each site separately and

then combine them together post-imputation. Our imputation models will include all covariates in the substantive analysis model described above, plus additional auxiliary variables to help predict missing values (e.g., other religious beliefs or practices about the focal deity, beliefs and practices regarding the other deity), which we assume will meet the Missing-At-Random assumption for unbiased imputations (White et al., 2011; Hughes et al., 2019; van Buuren, 2018). As the total amount of missing data is likely to be relatively small (estimated <10% per site), we will impute 10 datasets per society (this will be increased if we observed more missing data). These imputed datasets will be analysed using the "brm_multiple" command from the package "brms" (Bürkner, 2017), which pools results across each imputed dataset. We use multiple imputation rather than Bayesian imputation in Stan (McElreath, 2020) as multiple imputation is more flexible for handling different data types (e.g., categorical data), able to passively-imputed derived variables (i.e., punishment and omniscience mini-scales), and easier to incorporate auxiliary variables.

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